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DTCS–401

DIPLOMA (CS) (FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

COMPUTER ARCHITECTURE AND MAINTENANCE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain Computer Architecture with the help of a diagram. Also describe the use of Address Bus, Data Bus, and Control Bus. (CO1)

OR

(b) What are the different modes of operation of the CPU ? (CO1)

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2. (a) Draw the block diagram of North Bridge/South Bridge Architecture and describe the various blocks. (CO1)

OR

(b) Compare of the following : (CO1)

(i) PCI with PCI Express Bus

(ii) North-South Bridge/Hub Architecture

3. (a) What are the available storage devices ? Explain all the Recording Techniques. (CO1/CO2)

OR

(b) Write the important features of the following : (CO1/CO2)

(i) SDRAM

(ii) DDR

(iii) DDR2

(iv) DDR3

4. (a) What is Hard Drive Construction ? Explain its working with the help of a diagram. (CO1/CO2)

OR

(b) What is Cache Memory ? Describe L1, L2 and L3 Cache. Also discuss the performance of the cache memory. (CO1/CO2)

5. (a) What are the components of Motherboard ? State any eight motherboard selection criteria. (CO1 & CO2)

OR

(b) Write any *three* features of the below File Systems : (CO1 & CO2)

(i) FAT 16

(ii) FAT 32

(iii) NTFS

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